

Chapter 4 : Climate

Q.1. Make friends with maps! (Textbook page 25)

Study the following figures and answer the following questions in words/sentences:

Space for Brazil Climate Map (Textbook Page 25)



1. Considering the isotherms of Brazil, what is the average range of temperature in Brazil?

Answer: 28°C is the average range of temperature in Brazil.

2. In which area does it rain more in Brazil?

Answer: It rains more in the northern and central part of Brazil, e.g. Amazon Valley.

3. From which directions are winds flowing towards Brazil?

Answer: Northeast and Southeast direction winds blow towards Brazil.

4. What could be the reason behind that?

Answer: These are permanent trade winds which blow from the high-pressure belt towards the equatorial low-pressure belt. Since Brazil lies in the tropical zone, a low-pressure area is created and winds blow from the ocean towards the land.

5. What could be the obstruction in the way of these winds?

Answer: The Great Escarpment and the Brazilian Highlands could be the obstruction in the way of these winds.

6. Which type of rainfall will occur because of these winds?

Answer: Orographic rainfall occurs because of these winds.

7. Correlate these winds and rainfall.

Answer: The windward side of the Great Escarpment and the Brazilian Highlands receive abundant rainfall, while the rain shadow region is found on the other side.

8. In which part of Brazil are the average temperatures low?

Answer: The average temperatures are low in the Paraguay–Parana River Basin.

9. Identify the rain shadow area in Brazil. Describe its climatic characteristics.

Answer: The rain shadow region is found to the northeast of the Brazilian Highlands in Brazil. It receives only 0 mm to 500 mm of average annual rainfall.

10. In which region do you find a higher temperature?

Answer: We find a higher temperature in the region near the Equator in Brazil.

11. Considering the latitudinal extents of Brazil, where will you find temperate climate in Brazil?

Answer: We find temperate climate in Brazil to the south of the Tropic of Capricorn, i.e. south of the Great Escarpment and the Brazilian Highlands.

12. Describe the winds blowing between 0° to 5° N and S zone.

Answer: The winds blowing between 0° to 5° N blow from the northeast towards the Equator. The winds blowing between 0° to 5° S blow from the southeast towards the Equator.

13. Which method has been used to show distribution in the map given on page 25 of the textbook?

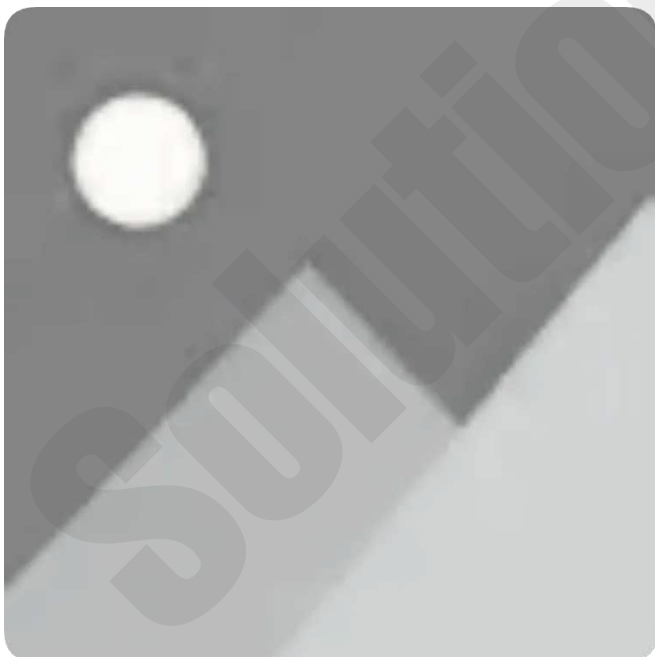
Answer: The isopleth method is used to show the distribution of annual average temperature and the choropleth method is used to show the distribution of annual average rainfall.

Q.2. Study the figures given in Q.1 and complete the following table:

Physiographic Divisions of Brazil	Annual Average Temperature in °C	Annual Average Rainfall in mm
Amazon Basin	28°C	2000 mm
Paraguay–Parana Basin	21°C	1500 mm
Brazilian Highlands	23°C	600 mm
Coastal areas	27°C	2000 mm

Q.3. Make friends with maps! (Textbook page 26)

Space for India's Climate Map (Textbook Page 25)



1. Which region gets more than 4000 mm of rainfall?

The windward side of the Western Ghats and the north-eastern region of India receive more than 4000 mm of rainfall.

2. Identify the regions with maximum and minimum temperatures.

The Thar Desert and the southern Peninsular region have the maximum temperatures, while the Himalayan region has the minimum temperatures.

3. In which direction is the temperature increasing?

The temperature increases from north to south.

4. Identify the directions of the winds shown. What are they known as?

The winds blow from the south-west towards the north-east. They are known as the South-West Monsoon Winds.

5. Which winds are responsible for the rainfall in India?

The South-West Monsoon Winds are responsible for the rainfall in India.

6. Some part of Rajasthan is under desert. What could be the reason for it?

Rajasthan receives very little rainfall because the Aravalli Range runs parallel to the monsoon winds, so they do not cause much rainfall in this region.

7. Which parallel of latitude passes through India affecting its climate?

The Tropic of Cancer passes through India and affects its climate.

8. In which part of peninsular India are semiarid climatic conditions found and why?

Semi-arid climatic conditions are found on the eastern side of the Western Ghats because it lies in the rain shadow region.

Q.4. Give it a try. (Textbook page 27)

Regions	Climatic Characteristics
Amazon Valley	Hot and humid throughout the year with heavy rainfall.
Highlands	Warm climate with moderate rainfall and comparatively lower temperatures.
Pantanal	Hot and humid with seasonal flooding.
Northern Coastal Region	Hot and humid with heavy rainfall.
Southern Coastal Region	Warm and humid with moderate rainfall.
Southernmost Region of Brazil	Mild climate with cool winters and moderate rainfall.

Q.5. Colours of Both (Textbook page 29)

Seasons	India	Brazil
Summer	March to May	December to March
Winter	December to February	June to August

Q.6. Observe the pictures given below and write a brief description about them.

Picture	Description
	(1) Traffic Jam due to Rainfall (India) Heavy rainfall causes waterlogging due to poor drainage.

As a result, roads get flooded and traffic jams occur, causing inconvenience to people.



(2) Traffic Jam due to Rainfall (India)

Flooded roads caused by heavy rainfall slow down transportation. Poor drainage creates severe traffic congestion in many cities.



Dry Land in Drought Affected Region (India)

The picture shows drought conditions due to very low rainfall. The land becomes dry, cracked and unsuitable for cultivation.



(4) Snowfall (India)

This picture shows snowfall in the Himalayan region during winter. Snowfall makes the climate extremely cold.



(5) Rainfall (Brazil)

The picture shows heavy rainfall in Brazil. Rainfall occurs frequently and keeps the climate warm and humid.



(6) Paddy Farming (India)

Paddy is cultivated in regions receiving abundant rainfall. It is one of the major crops grown in India.



(7) Deforestation (Brazil)

Trees are cut on a large scale for commercial purposes. Deforestation causes loss of biodiversity and environmental imbalance.



(8) Well – A Source of Water (India)

Wells are an important source of water in many rural areas. People use well water for their daily needs.



(9) Drought Quadrilateral Region (Brazil)

This picture shows a drought-prone region of Brazil where rainfall is very low. Farming becomes difficult due to dry conditions.